

Cough in Adults

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Cough is a sudden, forceful expulsion of air from the lungs. It is one of the most common reasons people see a doctor. The function of a cough is to clear material from the airways and to protect the lungs from particles that have been inhaled. People may cough on purpose (voluntarily) or spontaneously (involuntarily). (See also [Cough in Children](#).)

Coughs vary considerably. A cough may be characterized as dry (unproductive) or productive, bringing up blood or sputum (also called phlegm). Sputum is a mixture of mucus, debris, and cells expelled by the lungs. It may be clear, yellow, green, or streaked with blood.

People who cough very hard may strain their rib muscles or cartilage, causing pain in the chest, particularly when they breathe in, move, or cough again. Cough may be very distressing and interfere with sleep. However, if coughing increases slowly over decades, as it may in people who smoke, people may hardly be aware of it.

Causes of Cough

Cough occurs when the airways are irritated. Likely causes of cough depend on whether the cough has lasted less than 3 weeks (acute), for 3 to 8 weeks (subacute), or 8 weeks or longer (chronic).

Common causes

For **acute cough**, the most common causes are:

- An [upper respiratory infection](#) (URI), including [acute bronchitis](#)

- Inflammation and swelling of the mucous membrane of the nose ([rhinitis](#)) or sinuses (sinusitis) that causes postnasal drip (drainage of secretions from the nose down the throat, or pharynx)
- Flare-up of [chronic obstructive pulmonary disease](#) (COPD)
- [Pneumonia](#)
- [Pertussis](#)
- Flare-up of [heart failure](#)

For **subacute cough**, the most common cause is:

- Oversensitivity of the airways after resolution of a respiratory infection

For **chronic cough**, the most common causes are:

- Inflammation and swelling of the mucous membrane of the nose ([rhinitis](#)) or sinuses (sinusitis) that causes postnasal drip (drainage of secretions from the nose down the throat, or pharynx)
- [Gastroesophageal reflux](#)
- [Asthma](#)
- [Chronic bronchitis](#)
- [Chronic obstructive pulmonary disease](#) (COPD)
- Use of angiotensin-converting enzyme (ACE) inhibitors, which are blood pressure medications
- Cough that does not resolve and has no clear explanation

Less common causes

For **acute cough**, less common causes include:

- A blood clot in the lungs ([pulmonary embolism](#))
- A foreign object (such as a piece of food) that has been inhaled (aspirated)

However, people who accidentally inhale something typically know why they are coughing and tell their doctor unless they have dementia, stroke, or another disorder that causes difficulty with memory, cognition, or communication.

For **chronic cough**, less common causes include:

- [Lung cancer](#)
- [Tuberculosis](#)
- Fungal infections of the lungs

People who have dementia or stroke often have [trouble swallowing](#). As a result, they may aspirate small amounts of food and drink, saliva, or stomach contents into their windpipe (trachea). These people may repeatedly aspirate small amounts of these materials without their caregiver's knowledge and may then develop a chronic cough.

[Asthma](#) may cause cough. Rarely, the main symptom of asthma is cough rather than wheezing. This type of asthma is called cough-variant asthma.

The causes of [cough in children](#) are similar to those in adults, but [asthma](#) may be more common in children. Foreign body aspiration, uncommon in adults except in those with impaired development, dementia, or swallowing dysfunction, is more common in children.

Other causes include a psychogenic cough, which is a cough with no identifiable medical cause that is thought to be caused by psychological stresses. Some people may develop a psychogenic element in their cough even if there is a known medical cause (that is, they cough because they feel compelled to cough).

Evaluation of Cough

Not every cough requires immediate evaluation by a doctor. The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

In people with a cough, certain symptoms and characteristics are cause for concern. They include:

- Shortness of breath
- Coughing up blood
- Weight loss
- Fever that lasts longer than about 1 week
- Having risk factors for [tuberculosis](#), such as being exposed to tuberculosis, having [human immunodeficiency virus \(HIV\) infection](#), or taking steroids (sometimes called glucocorticoids or corticosteroids) or other medications that suppress the immune system
- Having risk factors for HIV infection, such as high-risk sexual activities or use of illicit drugs by injection

When to see a doctor

People who have warning signs should see a doctor right away unless the only warning sign is weight loss. Then, a short delay is not likely to be harmful. People who may have inhaled something should also see a doctor right away.

People with an acute cough but no warning signs can wait a few days to see whether the cough stops or becomes less severe, particularly if they also have a congested nose and sore throat, which suggest that the cause may be an upper respiratory infection (URI).

People who have had a chronic cough but no warning signs should see a doctor at some point, but a short delay is unlikely to be harmful.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause of the cough and the tests that need to be done (see table [Some Causes and Features of Cough](#)).

Some obvious findings are less helpful in making a diagnosis because they can occur in several disorders that cause cough. For example, whether sputum is yellow or green or thick or thin does not help distinguish bacterial infection from other possible causes. Wheezing may occur with [bronchitis](#), [asthma](#), or other disorders. A cough that brings up blood may occur in other conditions, including [bronchitis](#), [tuberculosis](#), and [lung cancer](#).



Some Causes and Features of Cough in Adults

Cause	Common Features*	Tests
Acute (lasting less than 3 weeks)		
A chronic obstructive pulmonary disease (COPD) flare-up	Wheezing, shortness of breath, and breathing through pursed lips Cough often produces sputum In people who already have COPD	Usually only a doctor's examination
A foreign object†	A cough that begins suddenly typically in people who have a disorder that interferes with communication, swallowing, or both, or in children No symptoms of an upper respiratory infection In people who otherwise are feeling well	A chest x-ray Bronchoscopy
Heart failure †	Shortness of breath that worsens while lying flat or that appears 1–2 hours after falling asleep Usually sounds suggesting fluid in the lungs, heard through a stethoscope	A chest x-ray Sometimes a blood test to measure a substance called brain natriuretic peptide (BNP) that is produced when the heart is strained

